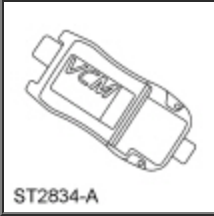


Daytime Running Lamps (DRL)



Special Tool(s)

 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
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Principles of Operation

The Body Control Module (BCM) controls the Daytime Running Lamps (DRL) by using various inputs to determine whether or not the DRL should be illuminated. These inputs include:

- Ignition status
- Headlamp switch position
- Headlamps on or off
- Transmission gear selection

If equipped with halogen headlamps, when the DRL is active, the BCM provides a pulse-width modulated voltage to the low beam headlamps. This illuminates the headlamps at a reduced intensity.

If equipped with HID headlamps, when the DRL is active, the BCM provides voltage to the front turn lamps.

The DRL is a programmable parameter for this vehicle.

Inspection and Verification

1. Verify the customer concern.
2. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
3. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

4. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

5. If the scan tool does not communicate with the vehicle:
 - verify the ignition is on.
 - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on).
If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
 - verify the scan tool operation with a known good vehicle.

- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

6. Check the operation of the PRNDL display by placing the gear selector lever in each of its positions.

- If the PRNDL display does not operate correctly, refer to Section 413-01.

7. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

8. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• The Daytime Running Lamps (DRL) are inoperative	• Exterior lighting system input/output • Transmission gear input • Body Control Module (BCM)	• GO to Pinpoint Test F.

Pinpoint Tests

Pinpoint Test F: The Daytime Running Lamps (DRL) Are Inoperative

Normal Operation

When the ignition is in RUN, the headlamps are not on (from the autolamp system or manual control from the headlamp switch), and the transmission is not in park, the Body Control Module (BCM) activates the DRL.

If equipped with halogen headlamps, when the DRL is active, the BCM provides a pulse-width modulated voltage to the low beam headlamps. This illuminates the headlamps at a reduced intensity.

If equipped with HID headlamps, when the DRL is active, the BCM provides voltage to the front turn lamps.

This pinpoint test is intended to diagnose the following:

- Exterior lighting system input/output
- Transmission gear input
- BCM

PINPOINT TEST F : THE DRL ARE INOPERATIVE

F1 VERIFY THE DRL OPERATION	
• Ignition ON. • Place the headlamp switch in the OFF position. • Place the gear selector lever in any gear other than PARK (P).	
Are the DRL illuminated?	
Yes	The system is operating correctly. INFORM the customer of the conditions required for the DRL to operate correctly.
No	GO to F2 .

F2 CHECK THE OPERATION OF THE HEADLAMPS

- Place the headlamp switch in the HEADLAMPS ON position and observe the headlamps.

Do the headlamps illuminate?

Yes	GO to F3 .
No	GO to Pinpoint Test A.

F3 CHECK THE TRANSMISSION GEAR INPUT

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Monitor the PCM transmission gear status PID (TR_F) while placing the gear selector lever through all its positions.

Does the PID agree with the gear selection?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	REFER to Section 307-01 to diagnose the transmission range input.